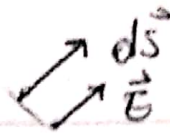


$$\rho = \frac{Q}{V} = \text{cte}$$

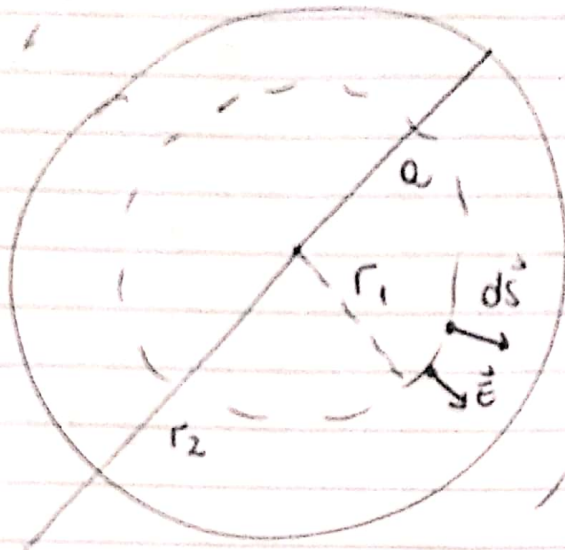
7:

a)



$$r_1 < a$$

$$r_2 > a$$



$$\vec{E} \parallel d\vec{S}$$

en todos
los casos

pues, $\rho > 0$

$$\rho V = Q$$

$$\rho V_{enc} = Q_{enc}$$

$$\oint \vec{E} \cdot d\vec{S} = \frac{Q_{enc}}{\epsilon_0}$$

$$\int E dS \cos 0 = \frac{\rho V_{enc}}{\epsilon_0}$$

$$E \int dS = \frac{\rho V_{enc}}{\epsilon_0}$$

$E = \text{cte}$ en
la sup de
la esfera gaussiana.

Aplicamos ahora a cada
caso:

a) Para $r_1 < a$:

$$E (4\pi r_1^2) = \frac{\rho \frac{4}{3}\pi r_1^3}{\epsilon_0}$$

$$E = \frac{\rho r_1}{3\epsilon_0}$$

$$\rho = \frac{Q}{\frac{4}{3}\pi a^3}$$

$$E = \frac{3Qr_1}{12\pi a^3 \epsilon_0} = \frac{Qr_1}{4\pi a^3 \epsilon_0} //$$

Encierra todo
la esta
carpeta.

b) Para $r_2 > a$

$$E(4\pi r_2^2) = \frac{\rho \left(\frac{4}{3} \pi a^3 \right)}{\epsilon_0}$$

$$E = \frac{3Q}{4\pi a^3} \cdot \frac{a^3}{3\epsilon_0 r_2^2}$$

$$E = \frac{Q}{4\pi \epsilon_0 r_2^2} //$$

c) Ahora $\rho = \rho_0 \frac{r}{a}$

$$\rho = \frac{dQ}{dV}$$

$$\therefore Q_{enc} = \int_{V_{enc}} \rho dV$$

no es cte

$$dV = 4\pi r^2 dr$$

$$\Rightarrow r_1 < a$$

$$Q_{enc} = \int_0^{r_1} \rho_0 \left(\frac{r}{a} \right) (4\pi r^2 dr)$$

$$Q_{enc} = \frac{4\pi \rho_0}{a} \int_0^{r_1} r^3 dr$$

$$= \frac{4\pi \rho_0}{a} \frac{r_1^4}{4}$$

$$= \frac{\pi \rho_0}{a} r_1^4$$

sólo hay carga
hasta $r=a$

$r_2 > a$:

$$Q_{enc} = \int_0^a \rho_0 \left(\frac{r}{a} \right) 4\pi r^2 dr$$

$$= \frac{\rho_0 4\pi}{a} \int_0^a r^3 dr$$

$$= \frac{4\pi \rho_0}{a} \frac{a^4}{4}$$

$$= \rho_0 \pi a^3$$

Así el campo eléctrico es:

$r_1 < a$:

$$E(4\pi r_1^2) = \frac{\frac{\rho_0}{a} r_1^4}{\epsilon_0}$$

$$E = \frac{\rho_0}{4a\epsilon_0} r_1^2 //$$

$r_2 > a$

$$E(4\pi r_2^2) = \frac{\rho_0 \pi a^3}{\epsilon_0}$$

$$E = \frac{\rho_0 a^3}{4\epsilon_0 r_2^2} //$$